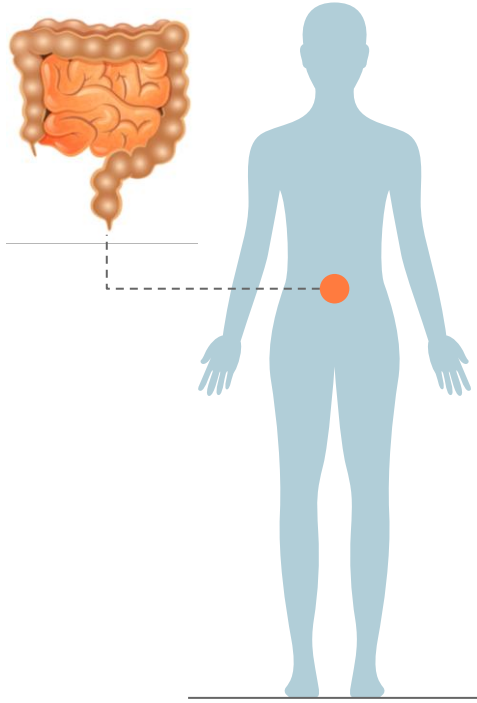
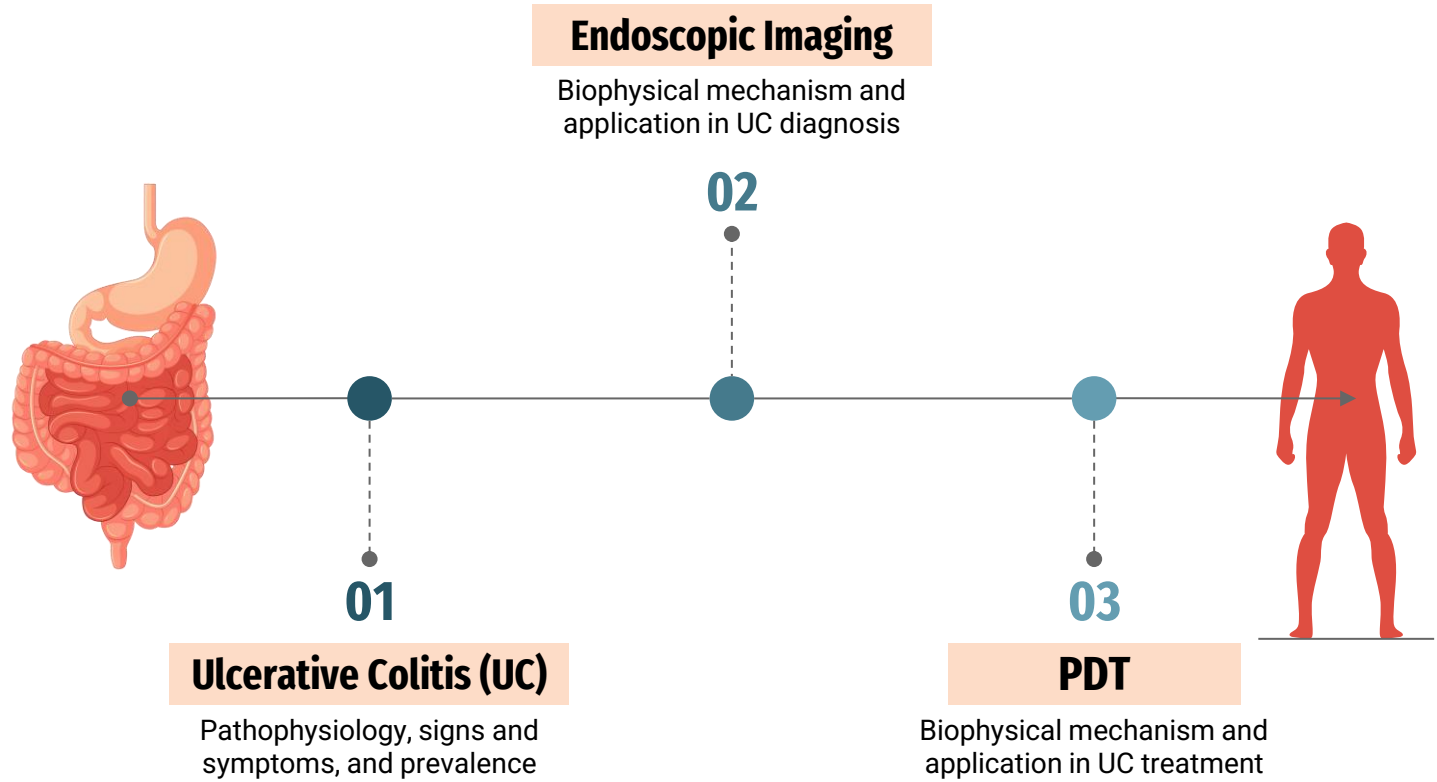




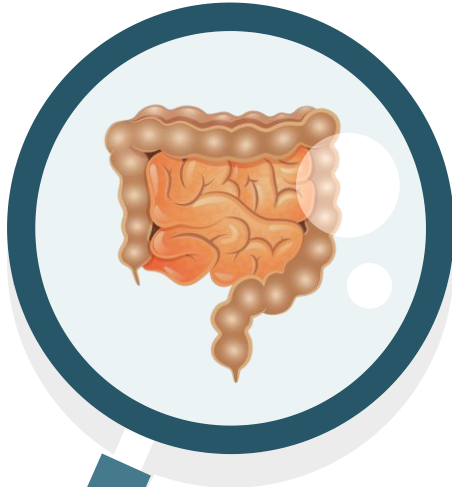
The Role of Endoscopic Imaging and Photodynamic Therapy in Managing Ulcerative Colitis

Joan Gabrielle Basa | PHYS 396 – Biophysics of Imaging

Overview of Content



Ulcerative Colitis: The Basics^[10-11,14,22]



- Most common type of **inflammatory bowel disease (IBD)** affecting the colon
- Inflammation in typically starts in the **rectum**
- Common symptoms of UC include **bloody diarrhea** and **abdominal discomfort**
- Characterized by periods of **flare-ups** and **remissions**
- Occurrence is **increasing globally**
- Goal of treatment is to heal the **lining of the colon**

Ulcerative Colitis: The Large Intestine (Colon)^[2,16,23]

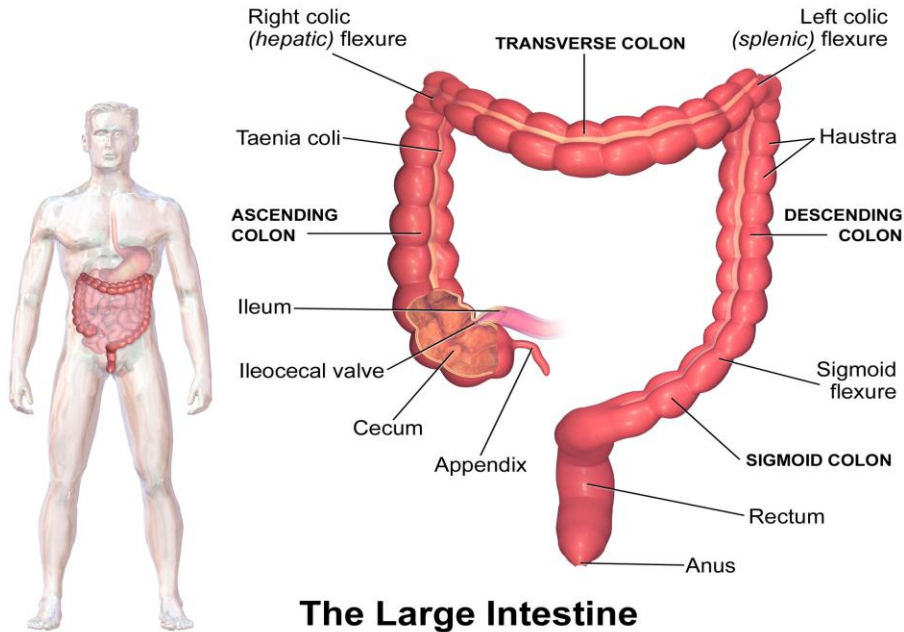


Figure 1: Anatomy of the large intestine^[2]

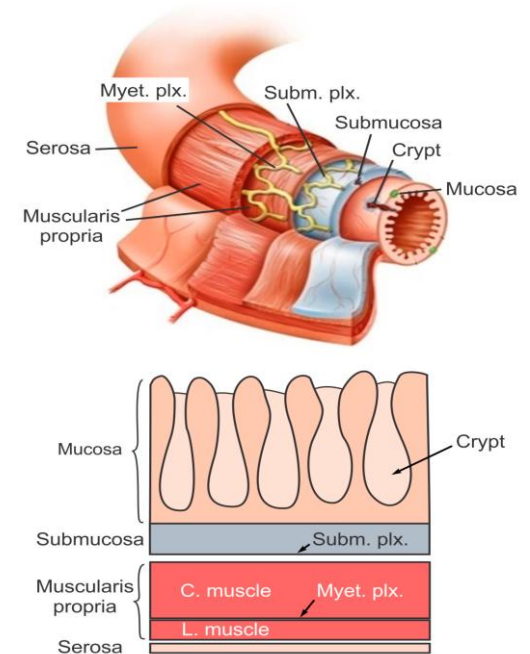
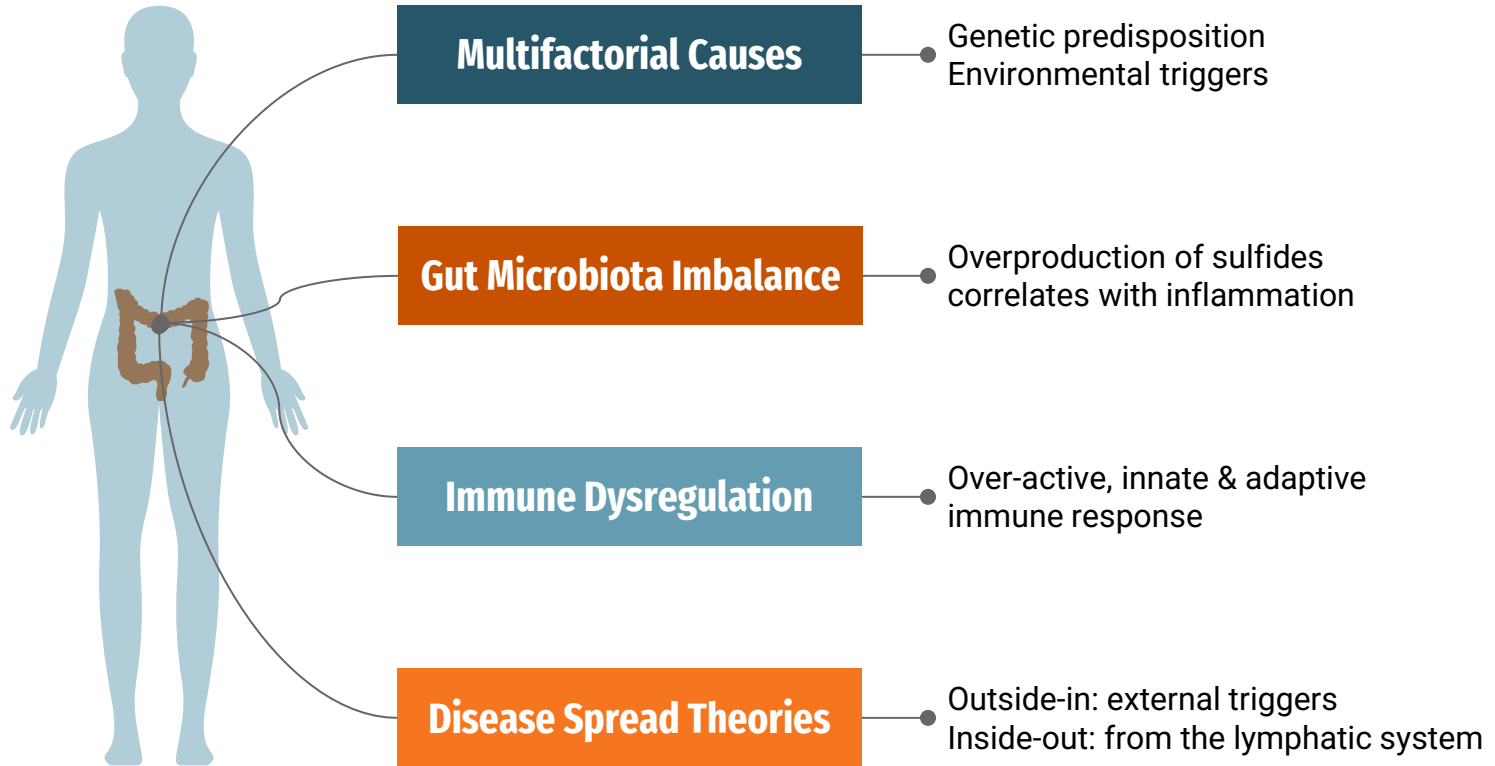
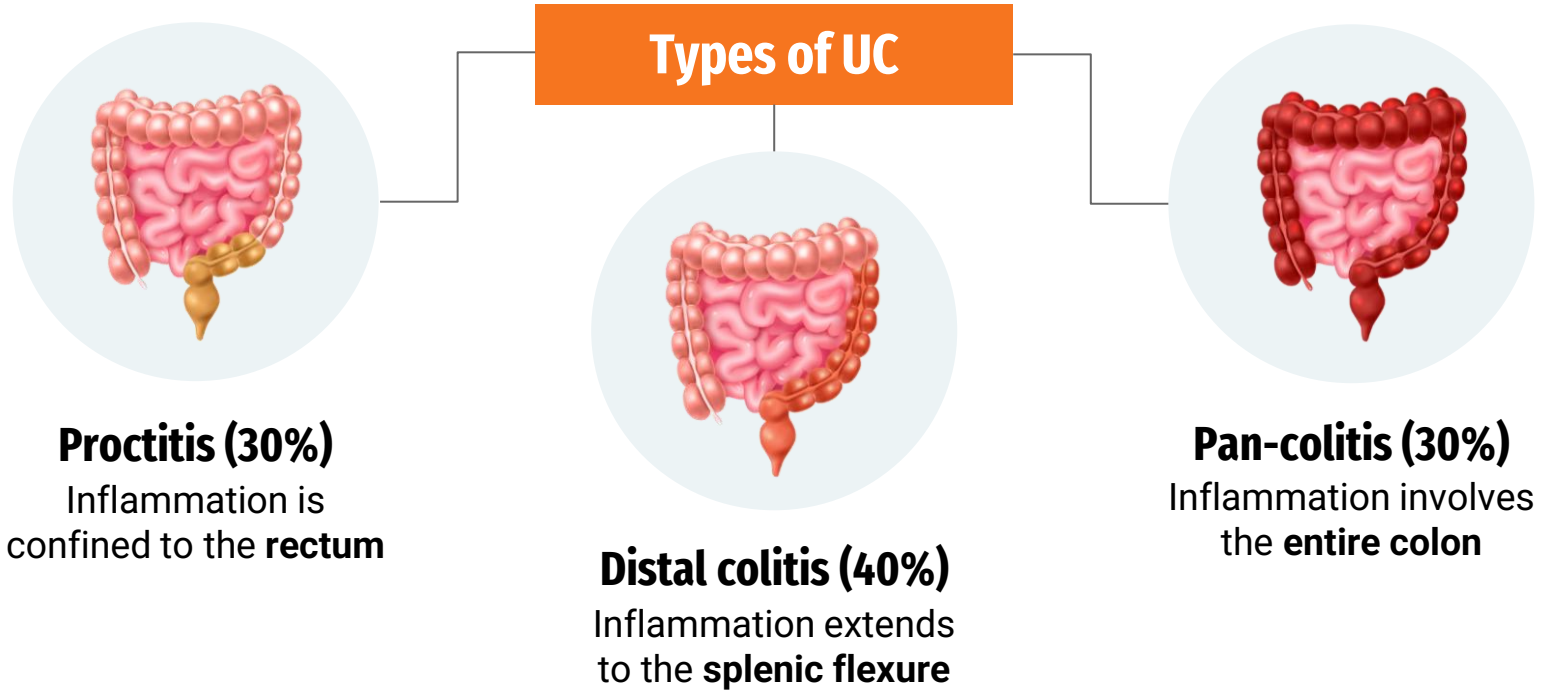


Figure 2: Layers of the colon lumen^[23]

Pathophysiology of Ulcerative Colitis^[10,16,22]



Classification of Ulcerative Colitis^[9]



Ulcerative Colitis: Diagnosis and Treatment^[9-11,14,16,22]



How it's
diagnosed



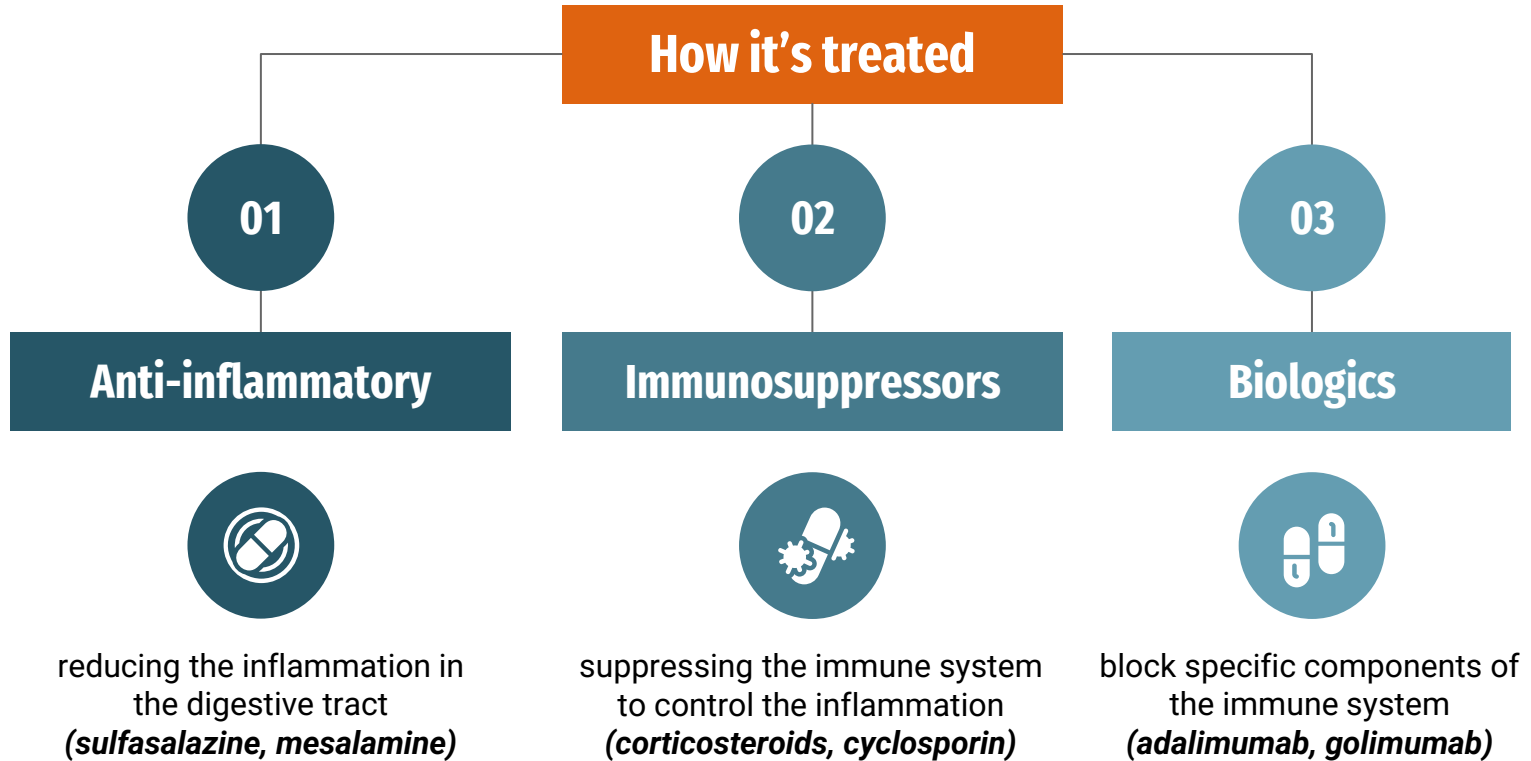
Colonoscopy

Allows for direct
visualization of the
colonic mucosa

Radiology

- CT scan
- MRI
- Ultrasound

Ulcerative Colitis: Diagnosis and Treatment^[16,22]



Endoscopy: Principles and Mechanisms^[1,13,22]

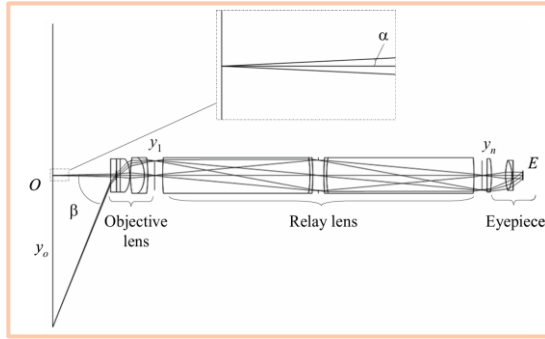


Figure 3: Schematic diagram of a typical endoscope optical system^[1]

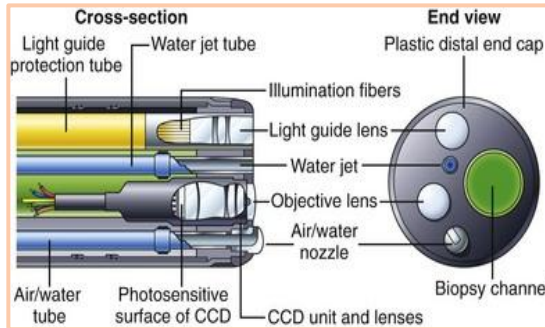


Figure 4: Distal tip assembly of a colonoscope^[1]

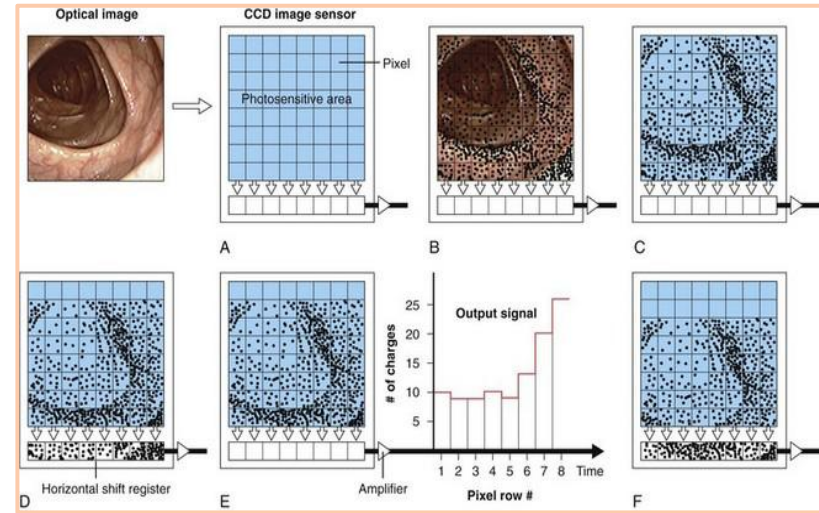
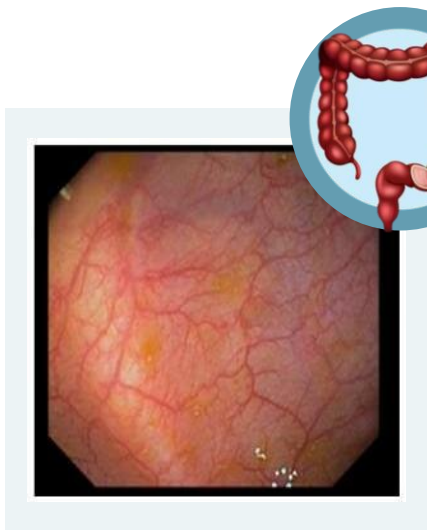


Figure 5: Image acquisition and readout using a line-transfer charge-coupled device (CCD)^[1]

- **Endoscope:** Flexible tube with light and camera
- **Types:** Upper endoscopy (mouth), colonoscopy (rectum)
- **Components:** Lens, light guides, air/water nozzle, biopsy channel
- **Image Processing:** Light brightens tissue, CCD captures image

Endoscopy: Application in UC^[8,22]



Normal

Smooth, shiny surface with visible blood vessels and light reflection



Mild to moderate

Rough, grainy texture with redness, reduced blood vessel visibility, and some erosions



Severe

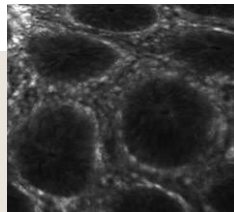
Spontaneous bleeding, severe redness, fragile tissue, and widespread mucosal damage

Endoscopy: Application in UC^[22]



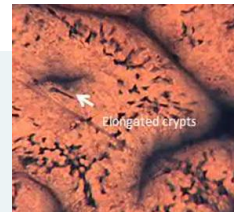
Chromoendoscopy

- Uses special dyes to highlight abnormalities in the gastrointestinal (GI) mucosa
- Helps detect dysplasia and cancer in conditions like ulcerative colitis (UC)



Confocal Laser Endomicroscopy

- Provides real-time, high-resolution imaging of cells inside the GI tract (cellular changes)
- Limited by high cost, specialized training needs, and small sampling area



Endocytoscopy

- Ultra-magnified imaging for cellular-level examination of tissues
- Used to analyze inflammation and detect precancerous changes

Endoscopy: Pros and Cons^[1,13,22]

PROS	CONS
Directly visualizes mucosal inflammation	Invasive procedure
Assesses disease extent and severity	Cannot assess transmural depth
Allows for tissue biopsies	Requires bowel preparation
Monitors treatment response	Can be uncomfortable for patients
Enables neoplasia surveillance	May have miss rates for dysplasia

PDT: Principles and Mechanisms^[3,5,12]

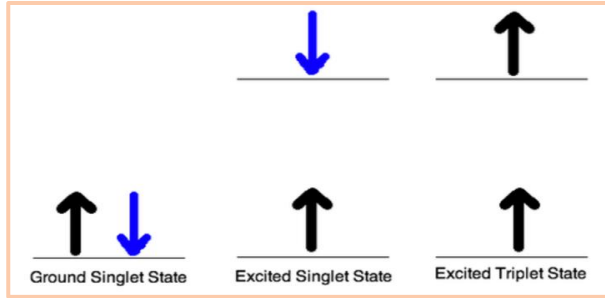


Figure 6: Photoexcitation from singlet to triplet state^[3]

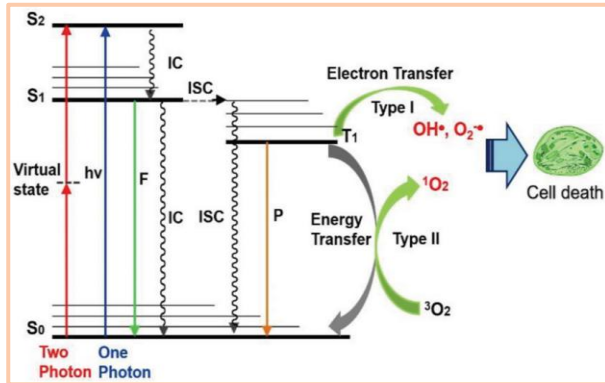


Figure 7: Production of ROS in Type I and II reactions^[3]

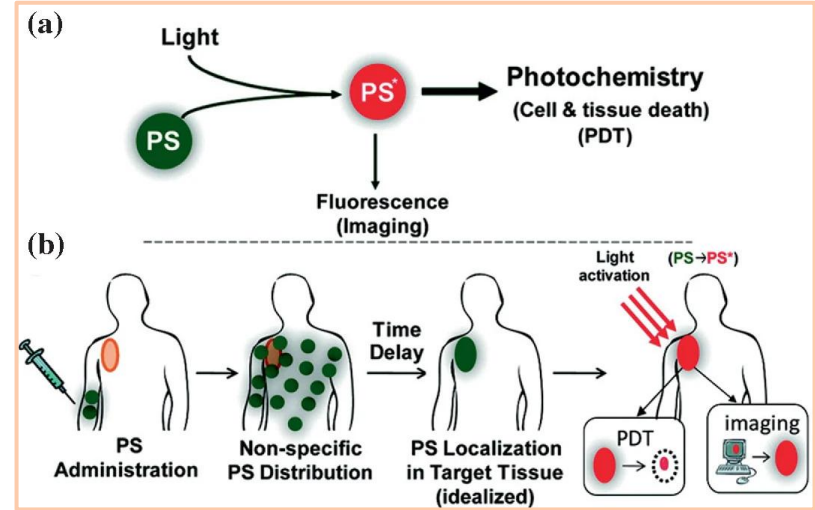


Figure 8: Schematic diagram of PDT action^[3]

- **Mechanism:** Light activates photosensitizers, boosting energy levels
- **ROS Production:** Radicals cause cellular damage
- **Cell Death:** Induces apoptosis or necrosis in target cells
- **Selective Targeting:** Photosensitizers accumulate in specific tissues

PDT: Application in UC^[3-4,18-21]

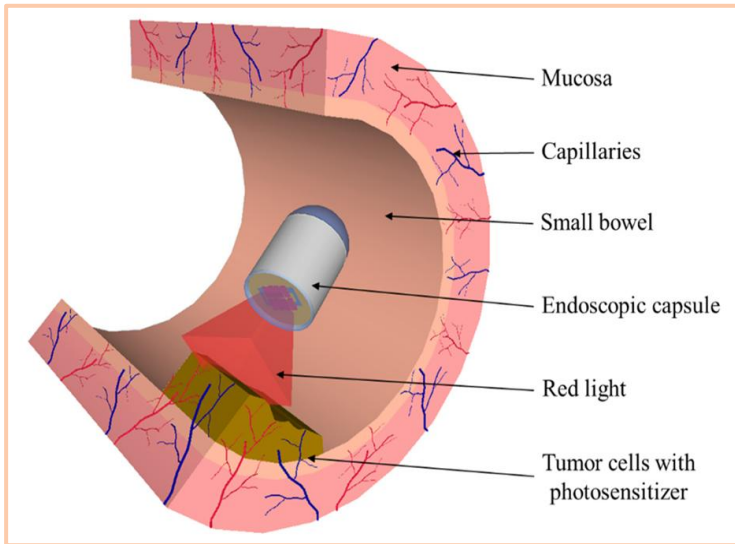


Figure 9: PDT-integrated endoscopic capsule^[3,19]

01

Anti-inflammatory
decreasing pro-inflammatory cytokines

02

Gut microbiota
help restore a more balanced composition

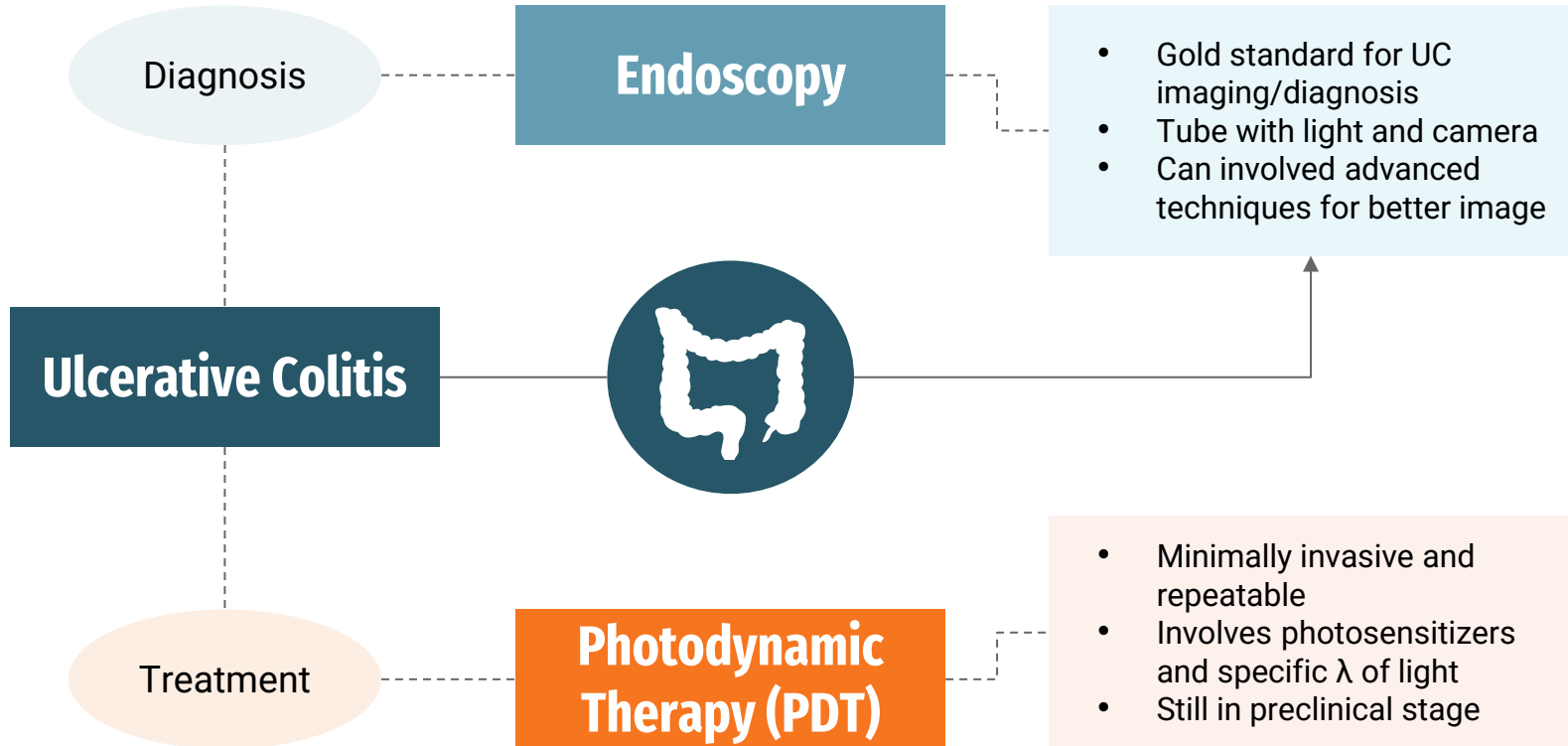
03

Minimally invasive
minimize damage to healthy tissues

PDT: Pros and Cons^[3,5,12,18-21]

PROS	CONS
Localized treatment	Limited light penetration in tissue
Minimally invasive	Dependence on oxygen and potential for hypoxia
Combinable and repeatable	Post-treatment photosensitivity
Potential for immunomodulation and anti-inflammatory effects	Photosensitizer properties and delivery challenges
Emerging evidence of efficacy in preclinical studies	Need for further clinical research and standardization

Summary



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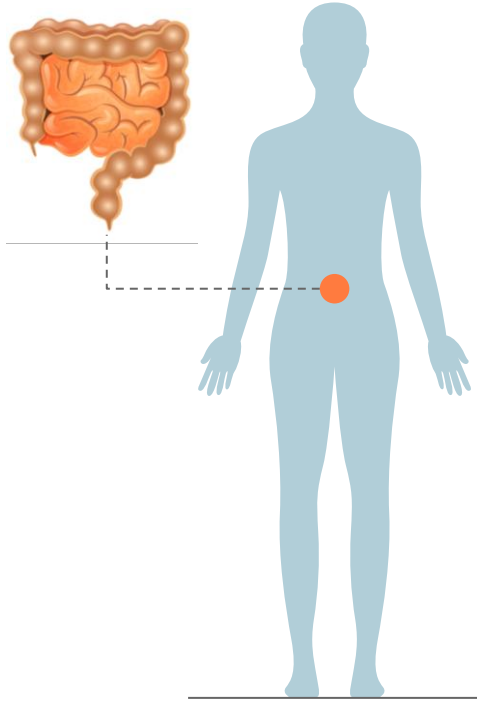
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